

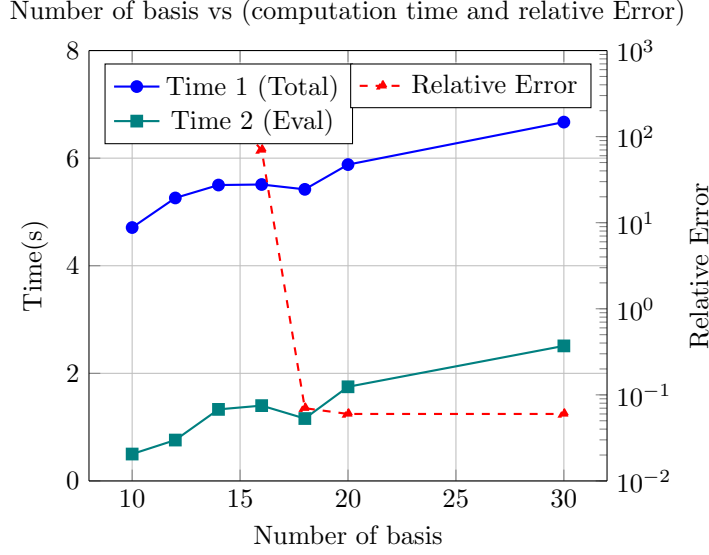


Figure 1: Relationship between the number of basis elements and computation time, and between the number of basis elements and relative error

1 Information on how the timings scale as the degree of the basis functions increases and on how the relative error descreases

The table shows the number of basis elements and computation time 1 (ODE loading time, computation time for the chebyshev polynomial and its zeros, computation time for $(L \bullet e_k)(t_j)$), computation time 2 (only computation time for $(L \bullet e_k)(t_j)$), and the maximum relative error with respect to the reference points $t = [3.8, 3.82, 3.84, 3.86, 3.88, 3.9, 3.92, 3.94, 3.96, 3.98, 4]$. The number of meshes for numerical integration is fixed at 200. Computation times were measured on a machine with an Intel(R) Core(TM) i7-10700K CPU @ 3.80GHz Risa/Asir 20260331. Unfortunately, due to page limitations, this data is not included in the paper, but it is put at <https://github.com/nobuki-takayama/sie-method-b>.

number of basis	time 1(s)	time 2(s)	relative error
10	4.71	0.50	250
12	5.26	0.76	250
14	5.50	1.33	250
16	5.51	1.40	70
18	5.42	1.16	0.07
20	5.88	1.75	0.06
30	6.67	2.51	0.06

Figure 1 is a graph of this table.